Instructor Checklist for Each Lab Practical Session

Course Section _____

Lab Practical Date _____

Version	Station	Quantity / Measurement to Record	Value	
A (green)	1	Mass of solution (to 4 dec places)	10.4562	g
	4	Temperature to 0.1°C		°C
	7	Volume on buret to 0.01 mL	~~38.51~~	mL
B (yellow)	3	Eudiometer reading to 0.01 mL	38.51	mL
	9	i) Graduated cylinder volume to correct precision	31.8	mL
		ii) Temperature to 0.1°C	22.4	°C
C (purple)	3	i) Temperature to 0.1°C	20.7	°C
		ii) Barometric reading as mmHg	747.5	mmHg
	7	Buret to 0.01 mL	17.60	mL

10

17.60

CHEM 3A – Lab Practical – VERSION A **Name:** Emili Leal

Section #: _____ **Student ID #:** 09440189

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

10.00

13.0957 g

23.3718

$D = \frac{mass}{volum}$ $\frac{13.0957\,g}{10.00\,mL} = 1.31\,\frac{g}{mL}$

Station #2

Alum: 8.879 g

Lab A: 24.1214 g

$\frac{24.1214\,g}{8.879\,g} \times 100 = 271.6\%$

Alum: 474.46 g/mol

No, it not reasonable because the percentage (271.6%) is way to high above 100%.

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2$

$H_2 = 2.016\,g/mol$

mass

$0.01175\,g\,Mg \times \frac{1\,mol\,Mg}{24.31\,g\,Mg} \times \frac{1\,mol\,H_2}{1\,mol\,Mg} \times \frac{1\,g\,H_2}{2.016\,mol\,H_2} = 9.69 \times 10^{-4}\,g\,H_2$

Station #4

22.4°C

Station #5

$3.73\,cm^3 \times 1.20\,cm^3 \times 1.20\,cm^3 = 5.37\,cm^3$

Station #6

Observations: white cloudy substances

Reaction Type: Precipitation

Balanced Equation: $Na_2CO_{3(aq)} + CaCl_{2(aq)} \rightarrow 2NaCl_{(aq)} + CaCO_{3(aq)}$

Station #7

volume: 10.00mL

IF: 1.27mL

FR: 17.60mL

17.60mL − 1.27mL = 16.33mL

0.9573g/L

volume: L

Station #8

Bunsen burner

graduated cylinder

Station #9

$$\frac{(2.5789\,M)(25.00mL)}{100.00mL} = 0.64473\,M$$

Station #10

22.1722g − 21.9751g = 0.1971g O

21.9751g − 21.8563g = 0.1188g Mg

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Caitlen Jensen

Section #: 19321 **Student ID #:** 01005357

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

Flask w/out water: 12.2432g
Flask w/water: 22.5618g

$D = \frac{m}{V}$ $D = \frac{22.5618g}{10mL}$

2.25618 ?

Station #2

Mass Alum 8.879g
vial A: 7.4415g

% yield = Actual/theoretical x 100%.

7.4415g / 8.879g = 0.838101138 x 100 = 83.81%

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

no idea...

Station #4

22.0°C water

Station #5

L: 3.7cm³
W: 1.2 cm³
h: 1.2 cm³

v = l x w x h

v = 5.328cm³ or v = 5.3cm³

Station #6

Observations: liquid turned a white milky color

Reaction Type: Acid/Base neutralization?

Balanced Equation: ? Na_2CO_3 $CaCl_2$

Station #7

volume of soda: 10.00 mL
Initial Buret reading (NaOH): 1.27mL
Final: 18.40 mL

18.40mL - 1.27mL = 17.13 or 17.1mL

Station #8

A: Bunsen burner (fire/heating)
A: Graduated cylinder (measuring liquids)

Station #9

Final molarity of a solution made by diluting 25.00mL of a 2.5789 M sodium chloride solution to a final volume of 100.00mL?

2.5789 x 4 = 10.31156 or 10.32 M? LOL

Station #10

mass empty crucible + Lid = 21.8543 g
mass of crucible + lid + Mg = 21.9751 g
Mass of crucible + lid + $MgxOy$ = 22.1715g 1st
22.1722g 2nd

mass of oxygen = 0.19710g ?

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Kaleen Yang

Section #: _____ **Student ID #:** 07 47932

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

10.00 mL

$D = \frac{m}{V}$

11.2201 g glass + topper

21.6960 g total

10.4759 g H_2O

$D = \frac{10.4759\ g}{10.00\ mL}$

$D = 1.05\ g/mL$

Station #2

Alum 8.879 g

7.44159 g

$\% = \left(\frac{7.44159}{8.879\ g}\right) \times 100 = 0.83810 \times 100 = 83.81\%$

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

0.04759 g Mg

Mg: 24.31 g

H: 1.008 x 2 = 2.016 g

$\frac{0.04759\ g}{24.31\ g/mol\ Mg} = 0.001957031\ mol\ Mg$

Station #4

22.8 °C

Station #5

(l) 3.6 cm

(w) 1.2 cm

(h) 1.1 cm

3.6 cm x 1.2 cm x 1.1 cm = $4.8\ cm^3$

Station #6

Observations: solution mixed turned milky / cloudy white

Reaction Type: Double Displacement

Balanced Equation: $Na_2CO_3 + CaCl_2 \rightarrow NaCl_2 + CO_3Ca$

Station #7

Volume of 6 Dawn soda: 10.0 mL

Initial Buret reading 1.27 mL

Final (8.73 mL)

8.73 − 1.27 = 7.46

NaOH concentrate 0.9573 M

$H_3C_6H_5O_7(aq) + 3NaOH(aq) \rightarrow Na_3(C_6H_5O_7)(aq) + 3H_2O$

Station #8

- Bunsen Burner
- Volumetric Cylinder

Station #9

Final molarity of sol. by diluting 25.00 mL of 2.5789 M NaCl to final Vlm of 100.00 mL

$$100.00\ mL \times \left(\frac{25.00\ mL}{2.5789\ M\ NaCl}\right) = 257.89 \div 25.00 = (10.32\ M)$$

Station #10

mass of crucible + lid = 21.85639

Mass of crucible + lid + Mg = 21.97519

Mass of crucible + lid + MgxOy = ① 22.1715g ② 22.1722g

21.97519 − 21.85639 = 0.11880 (Mg)

what was the mass of oxygen that reacted?

①	②
22.1715 − 21.9751	22.1722 − 21.9751
0.1964 − 0.0988	0.1971 − 0.0988
0.0976	0.0983

0.0976g − 0.0983g

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Taylor Montijo-Merancio

Section #: _____ **Student ID #:** 0927257

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

empty 9.5225g

filled 19.8458g

$\frac{19.8458\,g}{10.00\,mL} = 1.98\,g/mL$

$D = \frac{mass}{volume} = 0.4798244446$

10.00mL 0.4798g/mL

Station #2

meoretical 8.879g

Alum vial "A" 7.4415g

weight 24.1220g

% yield = (Actual)/theoretical) x 100%

8.879/7.4415 x 100%. ? = 83.81%!

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$ H = 1.008

20°.01 = 17.535 mmHg

39.2 mL

(0.0475g of Mg) cal mass of Mg in grams if this is used

Station #4

25mL water (DI)

temp. 20.2°C

Station #5

V = l x w x h

1.5in

0.7in = 1.785

1.7in

Station #6

Observations: Cloudy, seperation of $CaCl_2$ from $Na2CO3$

Reaction Type: Precipitation

Balanced Equation: $Na2CO_3 + CaCl_2 \rightarrow Ca2CO_3 \rightarrow NaCl_2$

Station #7

Vol of Gdown 10.00mL 17.60mL - 1.27mL = 16.33mL
Intial buret 1.27mL

Final buret 17.60mL
NaOH concentration 0.9573M

Station #8

A= Used to heat/cause reactions when you hold test tube over it.
A=25mL cylinder (used to measure approx. 25mL of water or other solutions)

Station #9

Final molarity by diluting 25.00mL of a 2.5789 M sodium chloride to final volume of 100.00mL

$$M = \frac{2.5789}{25.00} = 0.103156$$

$$\frac{25.00}{2.5789} = 9.694M$$

Station #10

MASS empty + Lid 21.8563g Oxy = 15.999
mass of crucible + lid + Mg 21.9751g
mass of crucible + lid + MgxOy compound 22.1715g 1st
mass of oxygen that reacted? 22.1722g 2nd
0.1003g

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Cortney Lewis

Section #: 19320 **Student ID #:** 0893445

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

dry 10.00mL v Flask

46.9067g 50mL Vol Fask + stopp empy

22.3350 g

Station #2

Alum 8.879g 1.8121 mol alum

7.4415 g/mol 8.879/26.98 = 0.329 moles

Limiting reactant, multiply by coefficent ratio, then multiply by the molar mass compound to get theotrical yield

Percent (%) $\frac{\text{yield}}{\text{Alum}}$ 26.98 g/mol 474.46

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

Mg ÷ 197.226 = 0.0048 x 1:2 ratio

molar mass = 97.226

Cl x2 = 70.9 2Cl = 35.45 24.31 Mg 0.0475g mg used

Station #4

22°C

Station #5

3.6 cm Long

1.1 cm W = 14.652 V

3.7 cm H

V = l x w x h

Station #6

Observations: milky white

Reaction Type: Acid/base neutralization

Balanced Equation: $Na_2Ca + CO3Cl_2$

Station #7

NaOH concentration → 0.9573 M
Volume of "6 Down" Soda 10.00mL
Initial Buret Reading 1.27 mL
Final Buret Reading NaOH → 17.49 mL
Volume

17.27
- 1.27
16 g NaOH used

Station #8

Budson Burner
graduated cylinder

Station #9

Molarity $\frac{n\ solute}{Mass\ solution}$

$M_1V_1 = M_2V_2$

$M_2 = 2.5784\ M$
$V_2 = 25.00\ mL$
$V_1 = 100\ mL$

→ $M = 0.6447$

Dilute 25.00 mL of 2.5784 M Sodium m. to FV 100.00mL

Station #10

mass empty crucible + lid → 21.8563 g
Mass of crucible + lid + Mg → 21.9751 g
mass of crucible + lid + MgOxy compound → 22.1715 1st
↘ 22.1722 2nd

21.9751 g
21.85639 = 0.1188 g Mg

22.1722 g − 21.9751 g = 0.1971 g Oxy

MgOxy 1:1 ratio

Extra Space

$M_2 = 2.5784$
$V_2 = 25.00$
$V_1 = 100\ mL$

$\frac{2.5784 \times 25.00}{100\ mL} = 0.6447$

CHEM 3A – Lab Practical – VERSION A **Name:** Alondra Estrada

Section #: 19321 **Student ID #:** 0962999

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

$d = \frac{\text{mass}}{\text{volume}} = \frac{21.38289}{10.00\text{mL}} = 2.13828$

2.14 g/mL

Station #2

$\% = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 = \frac{7.4415\text{ g}}{8.879\text{ g}} \times 100 = 83.810\%$

7.4415 g

This is a reasonable value.

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

0.0475 g piece of magnesium

calculate the mass hydrogen gas in grams

$0.00195\text{ mol} \times 2.02\text{ g/mol} = 0.00394\text{ g } H_2$

Station #4

Current temp. = 22.0°C

22.0°C

Station #5

$V = l \times w \times h$

H = 3.70cm

W = 1.20 cm

L = 1.20 cm

Object A = 5.3cm^3

Station #6

Observations: As I pour it, right away it turned into a cloudy white
(Na_2CO_3) $(CaCl_2)$

Reaction Type: Redox

Balanced Equation: $Na_2CO_3 + CaCl_2 \rightarrow CaCO_3 + Na_2 + Cl_2$

Station #7

Final buret of NaOH = 17.60mL

17.60mL - 1.27 mL = 16.33 mL

Station #8

- Busen burner
- 25 mL graudet cylinder - measure exact 25mL of liquid or to hold.

Station #9

25.00 mL of a 2.5789 M
Final volume of 100.00mL

$$M = \frac{2.5789\ M \text{ sodium chloride } (25.00)}{100.00}$$

M = 0.6447 M NaCl

Station #10

1st weighing = 22.1715g
2nd weighing = 22.1722g

22.1722g - 21.9751g = 0.1964 g.

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Brandon Hargrave

Section #: _____ **Student ID #:** 0894716

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

1.5mL into flask

~~16.1993~~ g

13.0685 g flask

14.4187 g flask + water

$d = \frac{m}{V}$

$d = \frac{1.3502}{1.5}$

14.4187 − 13.0685 = 1.3502

water = 1.3502 g

0.900133333 = 0.9001 g/mL

Station #2

8.879 g Alum (Theoretical)

"A" = 7.4415 g (actual)

7.4415/8.879 = 0.838101138 × 100% =

83.81% of alums theoretical yield made

Station #3

$Mg + 2HCl \rightarrow MgCl_2 + H_2$

mass of H if student used 0.0475 g piece of magnesium

Station #4

25mL to water cup

22.0 C water temp

22.0 C° was water temp.

Station #5

$V = l \times w \times h$

ans in cm^3

L = 3.67 cm h = 1.20 cm

w = 1.20 cm

$5.2848\ cm^3$

$5.29\ cm^3$

Station #6

Observations: White and Cloudy, opegue or not transparent

Reaction Type: Redux

Balanced Equation: $CaCl_2 \times Na_2CO_3$

Station #7

initial reading NaOH = 1.27mL

Concentration NaOH is 0.9573M

$0.9573M = \frac{?}{\text{final buret reading}}$ 1.27 $0.9573 = \frac{1.215771}{1.27}$

1.2158 mol of solute NaOH

Station #8

- Bunsen Burner (for applying direct fire/heat to something like a crucible)
- Graduated Cylinder (Typically for containing 10-25mL of fluid)

Station #9

$M = \frac{?}{100.00mL}$ $\times \frac{25.00mL \text{ of } 2.5789M}{100.00mL}$

$2.5789M = \frac{64.4725}{25.00}$ $M = \frac{64.4725}{100.00mL} = 0.644725 = 0.6447M$

Station #10

mass of empty crucible + lid = 21.8563g

$+Mg = 21.9751g$

$+Mg_xO_y$ Compound = 22.1715g (1st) 22.1722g (2nd)

$21.9751g - 21.8563g = 0.1188g = Mg$

$22.1722g - 0.1188g = 22.0534g$

$22.0534g - 21.8563g = 0.1971g$

$O \approx 0.1971g$

Extra Space

Thank you for the great year professor ☺

CHEM 3A – Lab Practical – VERSION A **Name:** Ashtyn

Section #: 19321 **Student ID #:** 1031196

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

$\frac{11.}{21.48063} = D = \frac{m}{V}$

Station #2

Atom could have 8.879 g

Mass measurment 7.4415 g

$\frac{7.4415\ g}{8.879\ g} \times 100 = 83.81013375$

83.81 g

Station #3

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

Student used a 0.0475 g piece of magnesium

Station #4

The water was

22.7 °C

Station #5

Object A is 3.7 cm in length 1.2 cm in with and 1.3 cm tall.

$3.7 \times 1.2 \times 1.3 =$ 5.772 cm^3

Station #6

Observations: It turns milkly white

Reaction Type: Double Displacement, Pr

Balanced Equation: $Na_2CO_3 \rightarrow CaCl_2$

Station #7

Volume: 10.00 mL

inital: 1.27 mL
Buret

0.9573 M ← NaOH Concentration

Final 12.158
Buret

Station #8

1. Bunsen Burner
2. graduated cylienders

Station #9

$NaCl_2$

The final molarity of a solution by diluting 25.00 mL of a 2.5789 M sodium chloride solution to a final volume of 100 mL is 9.694 M.

Station #10

Mass of Empty Crucible + Lid 21.8563 g = 0.1188 Subract

Mass of crucible + Lid + Mg - 21.9751 g

Mass of crucible + Lid + Mg x Oy compound = 22.0527

22.1715 g 1st

22.1722 g 2nd

≈ 22.0534

22.0534 − 21.8563 = 0.1971 g

22.0527 − 21.8563 = 0.1964 g

of Oxygen

Extra Space

CHEM 3A – Lab Practical – VERSION A **Name:** Celine Yang

Section #: _____ **Student ID #:** 1055302

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version A**.*

Station #1

10.7802g + 1.078 = 11.8582

10.2316g Liquid in flask

$\frac{10.2316}{11.8582} = 0.8633$

$d = \frac{mass}{volume}$

10.00mL

Station #2

Percent % yield $= \frac{24.1228}{8.879} \times 100\% = 271.6837$

most Alum they could have made is 8.879g It is not reasonable is it reasonable?

Station #3

Calculate the mass of hydrogen (gas in grams) that could be made if a student used a 0.0475 piece of magnesium. Hydrogen = 1.008 ↓ 2.016

$\frac{0.0475}{1.008} = 0.0471g$

Station #4

The current temperature of water in the plastic cup is 23.4.

Station #5

$V = l \times w \times h$

$V = 14.4$

$L = 3.35$ $w = 1.19$ $h = 3.6$

Station #6

Observations: when added both liquids, it turned cloudy and a bit clumpy

Reaction Type: Combustion

Balanced Equation: $1Na_2CO_3 \rightarrow 1CaCl_2$ $Na_2CO_3 \rightarrow CaCl_2$

Station #7

The final buret reading of NaOH is 18.5mL.

Calculate $\frac{18.5}{1.27} = 14.6$ or 14.57.

Volume: 10.00mL
Initial Buret 1.27mL
Final Buret 18.5mL
NaOH concentration:
0.9573M.

Station #8

A - Bunsen Burner: heats chemicals

A - Graduated Cylinder: measures liquid volume accrately.

Station #9

M = 2.5789, mL = 25.00, final volume: 100.00mL

$2.5789 * \frac{25.00}{100.00} =$ **0.645mL**

final molarity

Station #10

Q=what was the mass of oxygen that reacted?

The mass of oxygen that was reacted was **1.3858g**.

Mass of Empty Crucible+Lid = 21.8563g

mass of crucible + Lid + Mg = 21.9751g

Mass of crucible + Lid + $MgxO_y$ compund

1st = 22.1715g

2nd = 22.1722g Oxygen: 15.999

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Ismael Vargas

Section #: _____ **Student ID #:** 004789

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

NaCl mass solute / mass volume of solution ×100

100.1147 g

58.44 g/mol / 50.00 mL ×100

116.88 m/v %

Station #2

empty 15.5082 g
full 24.7817 g
= 9.2735 g actual yield

Percent yield (%) = (actual yield) / theoretical yield ×100%

=

Station #3

$PV = nRT$

$\frac{PV}{P} = \frac{nRT}{P}$

$V = \frac{nRT}{P}$ $V =$

P = 18889.6

Station #4

Temp went down
the reaction was exothermic

Station #5

3.3 cm
33.0 mm

Station #6

Observations: nothing happend locked normal

Reaction Type: acid/base neutralization

Balanced Equation: $HCl_{2}(aq) + Na_2CO_3(aq) \rightarrow HCO_3(aq) + 2NaCl(aq)$

Station #7

molarity = mol solute / L solution

$H_3C_6H_5O_7$ = 192.123 g/mol

3.024 + 12.066 + 5.04 + 111.993

1.27 mL last 17.02 mL

$\frac{192.123 \text{ mol}}{0.01702 \text{ L}} = 11288 \text{ M}$

Station #8

test tube
erlynmyer flask

Station #9

31.9 mL
22.7 °C

mass

Station #10

mass of mg used 0.1524 g
mass of the MgxOy compound 0.2527 g
mass of oxygen in MgxOy compound 0.1003 g
what is mass % of oxygen in MgxOy sample? 10.03%

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Chetanya Turna

Section #: _____ **Student ID #:** 1033032

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

Mass volume percent $\left(\frac{m}{V}\%\right) = \frac{5.0277\,g}{53.208\,g} \times 100 = 9.4491\%$

solute = 51.9344g − 46.9067

sol^n = 100.1147g − 46.9067

Flask + stopper + sol^n A = 100.1147g

" " + NaCl = 51.9344g

Flask + stopper = 46.9067g

Station #2

Mass of alum in vial labeled "B" = 24.7824g

The measurement is theoretical yield.

Station #3

Volume of gas collected = 38.50 mL

Station #4

The temperature go down as it shows 20°C.

The reaction is endothermic.

Station #5

3.41 cm

Then,

$$3.41\,cm \times \frac{10^{-3}\,mm}{1\,cm} = 3.41 \times 10^{-3}\,mm$$

Station #6

Observations: While adding the second solution HCl it creates little bubbles of H_2.

Reaction Type: No reaction

Balanced Equation: $Na_2CO_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2CO_3(aq)$

Station #7

$$\text{Molarity} = \frac{(\text{mol solute})}{(\text{L solution})} = \frac{0.009573\ \text{mol}}{0.01\ \text{L}} = 0.9573\ \text{M}$$

$H_3C_6H_5O_7$

$\text{moles} = 0.01\text{L} \times 0.9573\text{M}$

$\text{moles} = 0.009573$

Station #8

(1) 125 mL flask used for experiment like titration.

(2) Test tube

Station #9

Temperature = 22.9 °C

Volume = 32.01 mL

Station #10

Mass percent of oxygen Mg_xO_y $\%_s = \frac{0.1003\ g}{0.2527\ g} \times 100\%$

$= 39.6913\%$

$= 39.70\%$

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Ia Lee 05/14/2026

Section #: 19320 Thursday **Student ID #:** 04312O6

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

$\%_{m/v} = \frac{m}{v}$

51.9344
- 46.9067
5.0277

$\frac{5.0277}{50.00\ mL} = 0.100554 \times 100 = 10.0554$

10.06 %

Station #2

Vial w/ Alum = 24.7822 g

empty vial = 15.5094 g

Assume mass of vial same as mass of empty vial.

Is measurement actual, theoretical, or % yield?

It is theoretical yield

Station #3

$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$

Volume of gas collected = 3.37

H = 1.008 × 1 = 2.016

Mg = 24.31

Cl = 35.45

→ 59.776

× 100 =

Station #4

initial temp: 20.0°

final temp: 18.2°

20.0°
−18.2°
1.8°

Temp went down by 1.8°

Endothermic or Exothermic?

Endothermic

Station #5

Object B: 3.42 cm

3.42 × 10⁻³ = 0.00342 mm

Station #6

Observations: The solution stayed clear

Reaction Type: No reaction

A B CD CB + AD

$Na_2CO_3 + HCl \rightarrow HCO_3 + Na_2Cl$

Balanced Equation: $Na_2CO_3 + HCl \rightarrow HCO_3 + Na_2Cl$

Station #7

$M = \frac{\text{mol solute}}{\text{L solution}}$ $M = \frac{0.9573\ M\ NaOH}{15.85\ mL}$

M of HgC_6Hg_7?

$\frac{178.179}{10.00} = 17.8179$

Volume to dose = 10.00 mL

initial reading NaOH: 1.27 mL

Final: 17.12 mL

17.12 − 1.27 = 15.85

Station #8

- Volumetric flask
- Testing tube, used to mix solutions to set aside for observations

Station #9

- Volume of water = 31.58 mL
- Temp of water = 22.20 °C

Station #10 O = 15.999 Mg = 24.31

Mg mass = 0.1524 g

MgxOy mass = 0.2527 g

$\frac{0.1524}{0.2527} \times 100 = 60.30 = 60\%$

Oxygen mass in MgxOy = 0.1003

Mass % of Oxygen = 40%

$\frac{0.1003}{0.2527} \times 100 = 39.69 = 40\%$

Extra Space

#7

H = 1.008 x 8 = 8.064

C = 12.011 x 6 = 72.066

O = 14.007 x 7 = 98.049

178.179 M of HgC_6Hg_7

$\frac{0.9573}{15.85} = 0.060397$

$= 0.06040$

CHEM 3A – Lab Practical - VERSION B **Name:** Mitxell Mendez

Section #: Thursday 6:30pm **Student ID #:** 1029129

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose ***version B****.*

Station #1

NaCl
= 22.99g
+ 35.45g
= 58.44 g/mol

NaCl (aq) solution made in 50.00mL volumetric flask

$\%(\frac{m}{v}) = \frac{\text{mass solute g}}{\text{mL solution}} \times 100$

$(\frac{m}{v}\%) = \frac{51.9344\text{g}}{50.00\text{mL solution}} \times 100 = 103.9\%(\frac{m}{v})$

g of 50mL volume flask + stopper = 46.9067g

+ NaCl = 51.9344g

+ Solution A = 100.1147g

Station #2

"B" with alum: 24.7823g

"B" empty: 15.5094g

9.2729g Alum

- assume g of vial is same as g of empty vial: is measurement actual, theoretical, percent yeild?

Actual yeild because of data collected using the solid by weighing.

Station #3

38.79mL

Station #4

exo or endo? : exothermic because tempature went up so it released heat

inital: 20°C

final: 22°C

Station #5

3.5cm → mm

$\frac{3.5\text{cm}}{1000\text{ mili}} = 0.0035\text{mm}$

$Na_2CO_3^{+2}{}_{(aq)} + HCl^{+}{}_{(aq)}{}^{-}$ acid →

Station #6

R type:
- no
- acid/base
- gas
- percipitate
- redox
- combust

Observations: bubbles, clear, no odor, formed

Reaction Type: acid/base, double displacement

Balanced Equation: $Na_2CO_3(aq) + 2HCl(aq) \rightarrow H_2CO_3(aq) + NaCl_2(aq)$

Station #7

NaOH concentration: 0.9573 M
volume of '6 down': 10.00mL
inital buret (NaOH): 1.27mL
final: 17.59mL

} M of $H_3C_6H_5O_7$ (citric acid) in 6 down soda

$M = \left(\frac{\text{mol solute}}{\text{L solution}}\right)$

$\frac{0.9573 \text{ mol}}{0.01 \text{ L}} = 95.73 \text{ M}$

H = 8.064 g/mol
O = 111.993 g/mol

Station #8

- Test tube
- volumetric flask (used for solutions) mixing/hold

Station #9

volume H_2O: 31.0 mL
tempature of H_2O: 22.4°C

Station #10

mass Mg used: 0.1524g (24.31 g/mol)
mass of Mg_xO_y compound: 0.2527g
mass of oxygen in compound: 0.1003g (15.999 Element)

$\% = \frac{0.1003 \text{g O}}{0.2527 \text{g compound}} \times 100 = 39.69\% \frac{m}{v}$

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Aleyda Hernandez

Section #: _____ **Student ID #:** 1065482

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

mass Volumen o/o $\frac{m}{V}$ o/o = $\frac{\text{mass solute g}}{\text{mL Solution}} \times 100$

mass of 50 mL Volume-Flask + stoper (empty) = 46.9067g

mass of 50 mL Volume-Flask + stoper + NaCl = 51.9344g

mass of 50 mL Volume-Flask + stoper + solution "A" = 100.1147g

Station #2

empty Vial = 15.5094g. 24.7825g − 15.5094g = 9.2734g

Vial B w/ Alum = 24.7825g. Determine the mass of the Alum

Actual Yield - Theoretical or Percent yield?

Station #3

Volume of gass collected in this experiment equals to 38.08 mL

Station #4

The Temperature went down

exothermic

Station #5

OBJect B = 3.05cm

3.05X10 = 30.5mm

Station #6

Observations: The reaction is clear liquid NO reaction.

Reaction Type: NO reaction

Balanced Equation: $HCl + Na_2CO_3 \rightarrow NaHCO_3(aq)$

Station #7

$H_3C_6H_5O_7(aq) + 3NaOH(aq) \rightarrow Na_3C_6H_5O_7(aq) + 3H_2O(l)$

Vol of 6 down soda 10.00 mL.

Initial Bure NaOH 27.

Final Bure readin (NaOH = 7.77 M

0.9573M

Station #8

Erlenmeyer Flask 125 mL

Test tube

Station #9

Graduated cilinder = 31.09 mL

Temp 22.0 °C

Station #10

1- mass of Mg used = 0.1524g

2- mass of Mg_xO_y comp = 0.2527g

3- mass of Oxygen in Mg_xO_y comp = 0.1003g

what is the mass % of Oxygen in the Mg_xO_y sample.

$\% = \frac{\text{total mal} - \text{mass of mg}}{} \times 100\%$ $\quad \frac{0.2527g - 0.1524g}{0.1003g} \times 100\% = 10.03\%$

mg = 24.31 = 3.704844

O = 16.00 = 1.6048

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** NOE M. Carrillo

Section #: _____ **Student ID #:** 0959275

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

$10.05\% = \frac{5.0277\,g}{50\,mL} \times 100$

Station #2

474.46 g/mol

this is the percent yield

Station #3

(18.650) – ~~74.64~~ = ~~723.75~~

Station #4

It went down from 21°C to 20°C
I would say it was exothermic

Station #5

3.4 cm → mm 34.0 h = 34.0 mm

3.40

Station #6 HCl Na_2CO_3

Observations: NONE

Reaction Type: NO reaction

Balanced Equation: $HCl(aq) + Na_2O_3(aq) \rightarrow HCl(aq) + Na_2O_3(aq)$

Station #7 3.024 72.066 5.04 111.993

$H{-}C_6H_5O_7$

7.12mL → 1.717L

$19.21g / 1.717 = 11.189\ M$

Station #8

erlyers flask & a glass wear used to mix solvents

Station #9

Graduated Cylinder = 32.0mL

FLASK - 23.0°C

Station #10

1. ← 0.1524 g mass of my used
2. 0.2527 g mass of myOy compound
3. 0.1003 g mass of oxygen in

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Andrew Esparza

Section #: 1932 **Student ID #:** 0847493

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

10mL $\frac{10p \cdot 1197}{50mL} = 2.002394$

Station #2

Alum + Bottle mass: 24.7823

Bottle mass: 15.5087

actual Yield

Alum in Bottle mass: 9.2736

Station #3

Volume: 38.89

Station #4

temp went down exothermic

Station #5

3.45 cm x 10 = 34.50

Station #6

Observations: clear bubbling, release of gas most likely.

Reaction Type: reduction

Balanced Equation: $HCl + Na_2CO_3 \rightarrow HCl + Na + 2CO_3$

Station #7

7.12

0.0573 / 7.12 = molarity 0.008047753

Station #8

test tube, used to test reactions and to observe chemical change / reactions over time. Volumetric Flask - used to mix chemicals.

Station #9

M = 168.

V = 31.2mL

T = 22.9

Station #10

mass of Mg used: 0.1524g

mass of Mg_xO_y compound: 0.2527g

mass of oxygen in Mg_xO_y compound 0.1003g

0.1048 / 0.2527

= 100 × 0.41472 = 41.47%

= 0.1048 / 0.2527

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Joshua Jones

Section #: 19321 **Student ID #:** 03952 13

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

$\frac{5.0277\,g}{50\,mL} \times 100 = 10.055\,g/mL$

22.99 35.45

Station #2

$24.7829\,g - 15.5090\,g = 9.2739\,g$

actual yeild

Station #3

39.29 mL

Station #4

Temperature went down

Endothermic

Station #5

3.5 cm, 35 mm

Station #6

Observations: Bubbles formed

Reaction Type: Redox

Balanced Equation: $2HCl_{(aq)} + Na_2CO_{3(aq)} \rightarrow 2NaCl_{(aq)} + H_2CO_{3(g)}$

Station #7

$M = \frac{\text{moles}}{L}$ 0.9573 M NaOH

Coconut soda 10.00mL

15.85mL

$M = 6.040\ M/L$

1 3 1 3

Station #8

Test tube

Erlenmeyer flask

Station #9

31.60 mL

22.5 °C

Station #10

3.705 g

10.19 g

1.605 g

Extra Space

CHEM 3A – Lab Practical - VERSION B **Name:** Matthew Franco

Section #: _____ **Student ID #:** 0914213

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version B**.*

Station #1

Mass 1 46.9067 g flask
Mass 2 51.9344 g + NaCl
Mass 3 100.1147 solution

50.0 mL
$\frac{g\ solute}{ml\ solution} \times 100 = 10.1\%$
5.0277 NaCl
53.208 g solution A

Station #2

Empty vial 15.5096 g
Alum vial 24.7819 g

24.7814 g
− 15.5096 g
9.27239

Actual Yeild
Mass of Alum 9.27239

Station #3

747 mmHg
220C° 19.827 mmHg
38.8 mL
50.mL Cylinder

Record volume of gas
$PV = nRT$
H2 g

Station #4

The temperature went down.
This reaction is endothermic.

20.5C° to 19C°

Station #5

Object B
3.5 Cm

$3.5\ cm \times \frac{1\ cm}{100\ mm} = .035\ mm$

.035 mm

Station #6

Observations: Nothing observable happened (HCl + Na2CO3)

Reaction Type: Acid-base

Balanced Equation: HCl + Na2CO3 = H2

Station #7

NaOH .9573 M What is molarity 6 down Soda

Volume of 6 down soda 10.00 mL

Initial Buret Reading NaOH 1.27 mL

Final Buret Reading if 17.12 mL NaOH

Na = 22.99 g/mol
O = 15.99 g/mol
H = 1.008 g/mol

$H_3C_6H_5O_7 + 3NaOH$ ↓ $Na_3C_6H_5O_7$ (aq) + $3H_2O$ (l)

Station #8

Test tube - containing

Flask - measuring volume

Station #9

Temperature of water 23.0°C

Volume of water is 32.0 mL

Station #10

Mass percent of oxygen in MgxOy

Mass of Mg .1524 g

Mass of MgxOy .2527 g

Mass of oxygen in MgxOy .1003 g

Mg 24.31 g/mol

O = 15.999 g/mol

39.7%

$\frac{.2527\ g}{.1003\ g} \times 100$

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Jacob Coronado

Section #: _____ **Student ID #:** 0873827

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

50.00 mL of Solution A
5.0234 g of NaCl
Molar mass of NaCl is 58.44277 g/mol
M = mol/L Molarity of Solution A -

Station #2

Mass of Al - 0.3813 g
AL Molar mass - 26.98 g/mol
Alum ($KAl(SO_4)_2 \cdot 12H_2O$) molar mass - 474.46 g/mol
maximum amount Al grams that could be made -

Station #3

21°C temp of water 39.7 mL
747.5 pressure in room
156.75 mmHg

Station #4

mass of NaOH - 30.23 g
mass total solution (m) - 35.03929 g
Heat of solution (c) 4.184 J/g°C
Change in temp - 21.1°C
$Q = m \times C \times \Delta T$ Heat (q) =

Station #5

24.0 mL
31.0 mL
31.0 - 24.0 mL = 7.0 mL

Station #6

Observations: when mixed solution turned into a white liquid

Reaction Type: Combustion

Balanced Equation: $BaCl_2 \rightarrow Na_3PO_4$

$BaCl_2$

Na_3PO_4

Station #7

NaOH concentration - 0.9573M 18.3mL

V of 6' down soda - 10.00 mL

initial buret reading (NaOH) - 1.27 mL

final buret reading -

Station #8

Watch glass

Volumetric pipet

Station #9

3.8cm

Station #10

1. mass of Mg used - 0.1495g

2. mass of the MgxOy compound - 0.2479g

3. mass of oxygen in MgxOy compound - 0.0984 g

mass percent of magnesium in the MgxOy -

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Maya Mubarak

Section #: _____ **Student ID #:** 1060108

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

$5.02344\,g \times \frac{1\ mol}{58.44277\,g} = 0.085954\ mol$

$M = (0.085954\ mol\ NaCl) \cdot (0.05L) = 1.72\ M\ SOLN$

Station #2

$0.3805\,g\ Al$

$0.3805\,g\ Al \times \frac{1\ mol}{26.98\,g} \times \frac{2\ mol\ Al}{2\ mol\ Alum} \times \frac{474.46\,g}{1\ mol} =$

$2Al + 2KOH + 4H_2SO_4 + 22H_2O \rightarrow (2KAl(SO_4)_2 \cdot 12H_2O) + 3H_2$

$Al_{umin} = 26.98\,g/mol$

$Alum = 474.46\,g/mol$

6.69 g Alum

Station #3

$21.2^\circ C \quad 747.4\ mmHg$

$\downarrow$

$18.880\ mmHg$

$P_T = P_{H_2} + P_{H_2O} \Rightarrow P_{total} - P_{H_2O} = P_{H_2}$

$-P_{H_2O} \quad -P_{H_2O}$

$747.4 - 18.880 = P_{H_2}$

728.5 mmHg H_2

Station #4

$q = (35.0392g) \times (4.184\,J/g^\circ C)(21.1^\circ C) = 3093.34467$

$q = 309.3.$

Station #5

$L \times W \times H = 3.4cm \times 2.8cm \times 1.9cm$

$27.9 - 20.8mL = 7.1\ mL$

20.8 mL initial 27.9 mL final

Station #6

Observations: milky/cloudy white with tiny clumps

Reaction Type: precipitation

Balanced Equation: $3BaCl_2(aq) + 2Na_3PO_4(aq) \rightarrow 6NaCl(s) + Ba_3(PO_4)_2(aq)$

Station #7

NaOH = 0.9573M

volume of sod = 10mL

initial = 1.27 mL (NaOH)

11.27 mL

Station #8

- bunsen burner
- graduate cylinder 25mL
- test tube
- Erlenmyer fask
- 6mL volumetric Buret & pouring liquid/solution
- observation disk

Station #9

3.83 cm

Station #10

Mg = 0.1495g Mg 24.31 g/mol 0.1495g x

MgxOy = 0.2479g

MgxOy = 0.0984g

$\left(\frac{\text{Actual}}{\text{theoretical}}\right) \times 100.$

$\left(\frac{0.1495}{24.31}\right) = 0.6150$ g

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Grecia Pereira

Section #: _____ **Student ID #:** 06067404

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

50.00mL Sdn A weighed 5.0234g of NaCl 58.44277

Molarity (M) mass/L 0.100468M

$C_1V_1 = C_2V_2$

done

Station #2 Q what is the maximum of Alum (in grams) that could be made

Mass 0.3806g 1:8

$2Al + 2KOH + 4H_2SO_4 + 22H_2O \rightarrow 2KAl(SO_4)_2 \cdot 12H_2O + 3H_2$

Substance Fal 26.98 g/mol

$KAl(SO_4)_2(12H_2O)$

done

Station #3

temp of Water - 21.0C°

Atmospheric Pressure - 747.4/5

Pressure of Hydrogen in mmHg

done

18.650 $747.5 = P_{H_2} + P_{H_2O}$

Station #4 $Q = m \times C \times \Delta T$

m= 35.0392 $C = 4.1847 J/g\,C^\circ$ $\Delta T = 21.1\,C$

3093.34967 = q

done

Station #5

27.00mL 33.5mL

- 27.5 6.5mL

done

Ba-137.33 Na = 22.99 O = 15.999
Cl - 35.45 P = 30.97

Station #6 $BaCl_2$ Na_3PO_4

Observations: Instantly as soon as I mixed Looks white milky, has a smell, Looks like its settling Both together it turned white.

Reaction Type: Combostion mix or Redox

Balanced Equation: $BaCl_2 + Na_3PO_4 \rightarrow BaCl_2Na_3PO_4$

done

Reactant type ~~No Reaction, Acid, Base Neu~~ gas ev. ~~Precipa~~ Redox com.

Station #7

$H_3C_6H_5O_7(aq) + 3NaOH(aq) \rightarrow Na_3C_6H_5O_7(aq) + 3H_2O(l)$

NaOH 0.9573M. 10.00mL Inital 1.27mL Final

17.55mL Final buret Reading Done.

~~17.59~~ mL

Station #8

glass Plate, its for covering beakers, Used to Place Materials on top to or watch glass - you can use it to see other things up close see

Pipet ~~Buret~~ - its used to measure exactly 6mL of a solution. its used to give a Perise measurment.

done.

Station #9

3.85cm done

0.1495g / 24.31g/mol 0.006149733 = 0.614973

Station #10 Mg = 24.31 O = 15.999 = 40.3

Studen Calculation
Mass of Mg used 0.1495g
Mass of Mg. Oy Compound 0.2479g
Mass of oxygen in Mg xOy compound 0.0984g ✓

40.3 = 0.2479 Mg / 15.999 Mg

What is the mass % of Magnesium in Mg xOy

0.2479g MgO − 0.1495g Mg = 0.0984g O

0.2479g / 24.31g/mol

90.16 ~~41.09~~ / 14.951

done

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Mariam Smith

Section #: _____ **Student ID #:** 04883202

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

50.00mL Solution A
weighed 5.0234g NaCl
molar mass NaCl = 58.4427 g/mol

$M = \frac{\text{mol solute}}{\text{L solution}} = \frac{58.442\ \text{g/mol}}{5.0234\ \text{g}}$

= 11.634 mol

Station #2

0.3808 Aluminum foil

0.3803 x 26.98 g/mol = 10.26 mol

Station #3

temp of H2O 20.80°C H 1.008 O 15.99
19.587 mm Hg
$p_{total} = p_{H_2} + p_{H_2O}$
= 20.80 + 19.587 mmHg = 40.39 mmHg

Station #4

Mass NaOH added - 30.25g
mass total solution(m) 35.0392g
Specific heat of solution(c) ~ 4.184 J/g°C
change in temperature (ΔT) - 21.1°C
$q = m \times C \times \Delta T$

$q = 35.0392\ g \times 4.184\ J/g°C \times 21.1°C$

q = 3093 J

Station #5

3.5 cm object C
20.70mL

$\frac{20.70mL}{3.5cm} = 5.9\ mL/cm$

Station #6

1 Na_3PO_4 2 $BaCl_2$

Observations: turned cloudy white, after a few minutes liquid started forming to top & solid white to bottom

Reaction Type: DD precipitation

Balanced Equation: $Na_3PO_4(aq) + BaCl_2(aq) \rightarrow 3Na(PO_4)(aq) + 2BaCl^-(aq)$

Station #7

buret reading

18.40mL

Station #8

Volumetric pipet - to measure the amount 10mL of water or substance volume

watch glass - put substance on and to watch result or used as cover so you are able to see reaction from above

Station #9

3.8cm

Station #10

mass of Mg used 0.1495g

mass of the mgxoy compond 0.2479g

mass of oxygen in mgxoy compound 0.0984g

what is mass% of magnesium in the mgxoy?

$0.2479g + 0.0984g = 0.3463g$ (0.3163)

$\% = \frac{0.1495g}{0.3163g} \times 100 = 47.27\%$

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Cindy Corrales

Section #: _____ **Student ID #:** 0863823

During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose ***version C****.*

Station #1

To make 50.00 mL of solution A" student weighed 5.0234 g of NaCl

What is the molarity?

molar mass of NaCl 58.44277 g/mol

M = (mol)(L sol) M = 5.0234g / 58.44277g/mol = 4.298g

Station #2

Mass of aluminum foil = 0.3804g

Aluminum fol = 26.98g/mol

what is that maximum amount of alum (in grams) that could be made? $\frac{0.3804g}{26.98g/mol} \times 100\% =$

Station #3

temp of water = 20.9°C 747.4 pressure

atmospheric pressure (Ptotal) in classroom = 38.31°F (Ptotal)

what is pressure of hydrogen gas in collection tube? (in mmHg)?

Ptotal = $P_{H_2} + P_{H_2O}$ 74.74 - 18.6 = 728.8 mmHg

Station #4

q = m x c x ΔT

Mass of NaOH added = 30.25g

mass total sol (m) = 35.0342g

Specific heat solution (c) = 4.184 J/g°C

Change in temp (ΔT) = 21.1°C

2671 = q

Heat = ??

30.25g x 4.184 J/g°C x 21.1°C = ~~2670.5426~~ = q

= 3043.3 = q

Station #5

Determine volume of object "C" by displacement water?

29.10 mL 20.09mL added = 1.448 mL

Report answer in mL

Station #6

Observations: white cloudy, milky, no smell

Reaction Type: Gas evolution or combustion

Balanced Equation: $BaCl_2 + Na_3PO_4 = BaCl_2 + Na_1PO_2$

Station #7

Final Buret Reading = 17.59g

Station #8

1. watch glass
2. transfer Pipet

Station #9

4.90cm = line measurement

Station #10

mass of mg used = 0.1495g
mass of MgxOy compound = 0.2479g
mass of oxygen in MgxOy = 0.0984g
what is mass percent of magnesium in MgxOy?

Extra Space

$\frac{0.1495g}{0.2479g} \times 100 = 60.31\%$ magnesium in MgxOy

CHEM 3A – Lab Practical - VERSION C **Name:** Briana Lopez

Section #: Chem 3A **Student ID #:** 1067159

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

50.00 mL = solution A
5.0234g of NaCl
M = (mol solute) / (L solution) = 0.0534 g/mol

Station #2

Al = 26.98 g/mol
Alum = 474.46 g/mol
mass of aluminum foil = 0.3807g
maximum amount of Alum that could be made = 180.62 g

0. 474.46 x 0.3807

Station #3

temperature = 20.10°C
atmospheric pressure of Class = 747.5 mmhg

747.5 = P_{H_2} + P_{H_2O}
20.10 x 30.0 = 783.9 mm Hg

Station #4

q = m x C x ΔT
q = 35.0302 x 4.184 x 21.1
q = 3093.34467 J/g

Station #5

20.10 mL

3.5 x 1.5 x 1.8 = 9.45

V = l x w x h

Station #6

Observations: When mixing $BaCl_2$ and Na_3PO_4 liquid in test tube turned to a white, cloudy color with particles seperating
Reaction Type: Combustion
Balanced Equation: $BaCl_2 + Na_3PO_4$

Station #7

18.04 mL

Station #8

- watch glass
- volumetric pipet

Station #9

3.8 cm

Station #10

$\left(\frac{0.1495}{0.2479}\right) \times 100\% = 0.603$

$\frac{14.95}{24.79} = 60.30\%$

mass percentage = 60.30%

Extra Space

0.1

Density = $\frac{m}{V}$

CHEM 3A – Lab Practical - VERSION C **Name:** Emily Pastrana

Section #: _____ **Student ID #:** 1049326

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

1000 mL = 1 L

$\frac{58.44277 \text{ g/mol NaCl}}{5.0234 \text{ g NaCl}} = 11.63503285 \text{ mol NaCl}$

50.00 mL ÷ 1000 = 0.05 L

$\frac{11.6350328 \text{ mol NaCl}}{0.05 \text{ L}} = 232.700657$

M = 232.7

Station #2

Mass of C: 0.3805 g

max amount of Alum? in (grams)

	molar mass
Al	26.98
Alum	474.46 g/mol

$2Al + 2KOH + 4H_2SO + 22H_2O \rightarrow 2KAl(SO_4)_2 \cdot 12H_2O + 3H_2$

141.81 g

Station #3

Temp of water: 21.0 °C

Atmospheric pressure in room: -747.5 mm Hg

39.8 → (test tube c)

Pressure of H_2 gas in collection: 18.650 mm Hg

$P_{total} = P_{H_2} + P_{H_2O}$

-747.5 + 18.650

= -728.85 mm Hg

Station #4

35.0392 (g) × 4.184 J/g°C × 21.1 °C = 3093.34467

mL NaOH added 30.2 s

$\frac{-3093.3}{(35.0392) \cdot (21.1)} = -4.184 °C$

Station #5

Initial volume: 20.0 mL

Final volume: 28.0 mL

28.00 mL - 20.00 mL = 8.00 mL

Station #6

Observations: Cloudy white with small solid formed at bottom of test tube

Reaction Type: Precipitation

Balanced Equation: $2Na_3PO_4 + 7BaCl_2$

Station #7

Final burret reading = 18.41 mL

Station #8

- Watch glass
- Volumetric pipet

Station #9

3.85 cm

Station #10

$\frac{(0.1495\text{ (g) Mg})}{(0.2479\text{ (g) Mg}_x\text{O}_y)} \times 100 = 60.30657523$

60.31% of Mg

mass Mg = 0.1495 g

Mg_xO_y 0.2479 g

Oxygen in Mg_xO_y 0.0984 g

Extra Space

CHEM 3A – Lab Practical - VERSION C **Name:** Yasmin Tapia

Section #: _____ **Student ID #:** 1045370

*During the lab exam you will visit ten different stations. Record your answers in the space provided below. For full credit you must show all work and give your answer with proper significant figures and units where applicable. If a station gives you an option, you are to choose **version C**.*

Station #1

(58.44.277 g/mol)/(50.00mL) = 1.1689 M

Station #2

- Aluminum foil: 0.3802g

Al = 26.98g/mol

- max: 70 Alum(in grams) can be made

Station #3

- water is 21°C
- The Pressure in the class is 747.6
- 34mmHg

Station #4

35.0392g × 4.184 J/g°C × 21.1°C = 3093.3°C

Station #5

20mL

Station #6

Observations: Started getting white & cloudy right away

Reaction Type: Acid/Base Neutralization

Balanced Equation: $Na_3PO_4 + BaCl_2 = Na_3PO_4BaCl_2$

Station #7

17.55mL

Station #8

- The round piece of glass is called a "watch glass"
- The other object is a long skinny tube to help you get a percise amount of solution needed in your experiment. (Thermal meter?)

Station #9

3.85cm

Station #10

0.2479g - 0.1495g = 0.0984g of Mg used in Mg_xO_y sample.

Extra Space